DocuMind Academic Benchmark: Distributed Consensus

Abstract:
Fault-tolerant distributed state machine replication relies on consensus protocols. We evaluate the Raft consens

1. System Model and Assumptions:
We assume a partially synchronous network model where messages may experience arbitrary bounded delays.

2. Raft Consensus Protocol:

Raft organizes consensus through leader election, log replication, and safety invariants. Nodes exist in one of thr

3. Performance and CAP Trade-offs:
Under network partitions, Raft prioritizes consistency over availability (CP system under CAP theorem). In our 5-